

MiSeq Loading

A step-by-step guide





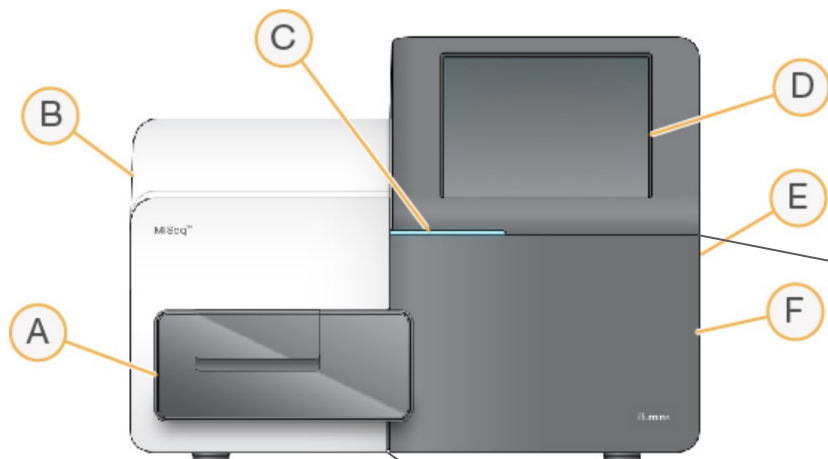
4-channel system (4 dyes)

- 1 - 24 Million reads
- 2 x 300 bp
- Max 600 cycles
- Run 4-55 hrs

4-Channel Chemistry				
	 A	 G	 T	 C
Image 1				
Image 2				
Image 3				
Image 4				
Result	A	G	T	C

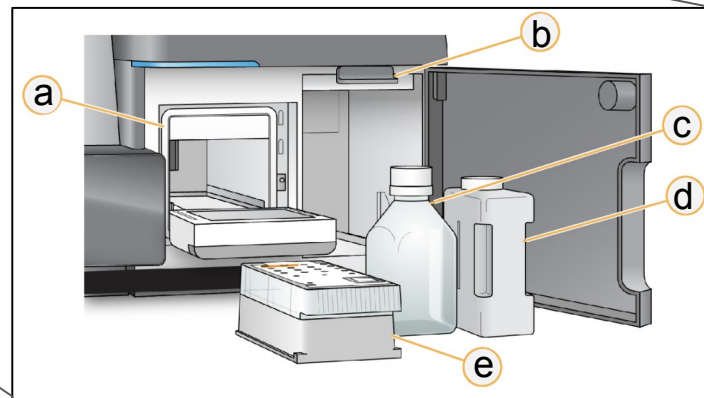
- Sequencer takes four images per cycle
- **Applications:** small whole genomes (microbiome), targeted genes, 16S metagenomics
- Handles low diversity runs well

Instrument Components

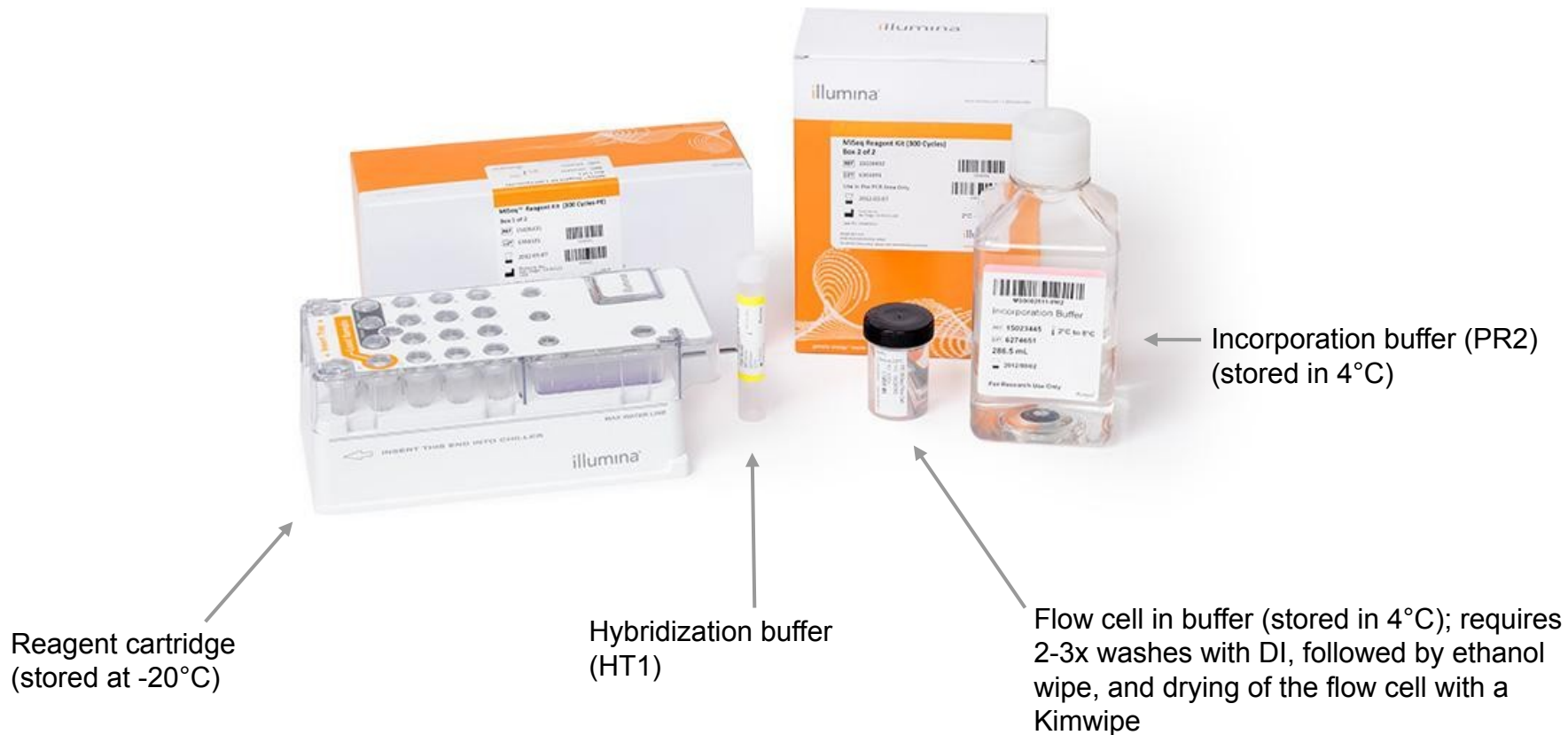


- A. Flow cell compartment
- B. Enclosed optics compartment
- C. Status bar
- D. Touch screen monitor
- E. External USB ports
- F. Reagent compartment

- a. Reagent chiller
- b. Sipper handle
- c. PR2 bottle
- d. Waste bottle
- e. Reagent cartridge



MiSeq Kit

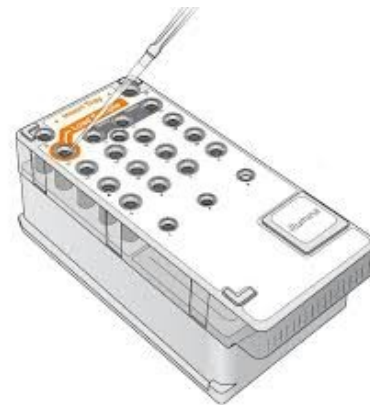


Sequencing Kits and Loading Concentrations

- MiSeq V2 (V2 - 12M, Micro- 4M, Nano-1M)
 - Available in 50, 300, 500 cycles kit
 - Our optimized loading concentration: 10pM in 600 μ L
 - Cluster Density Goal: 800-1000/mm²
- MiSeq V3 (24M)
 - Available in 150, 600 cycles kit
 - Our optimized loading concentration: 12pM in 600 μ L
 - Cluster Density Goal: 1000-1200/mm²

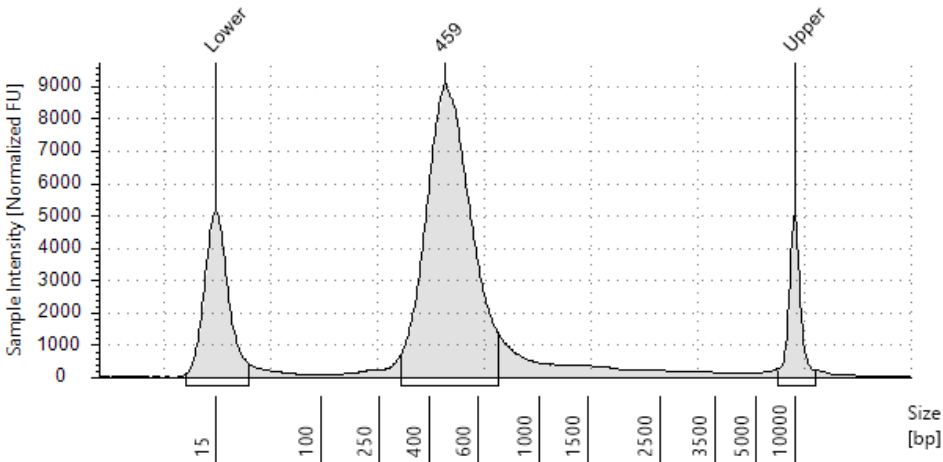
Before Loading

- Thaw Reagent Cartridge
 - Overnight in 4°C or water bath (room temp ~1 hr)
- Take out Flow Cell and Incorporation Buffer from 4°C
 - At least 30 minutes before loading to equilibrate to room temperature
- Invert the cartridge 5 times to mix - avoid making bubbles
- Carefully pierce foil (well #21) with a pipette tip once ready to load the diluted library into cartridge



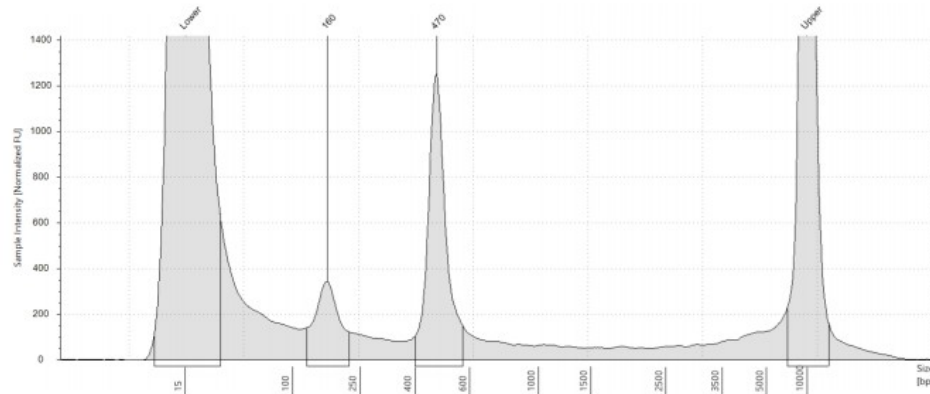
Library Distribution

Good Sample

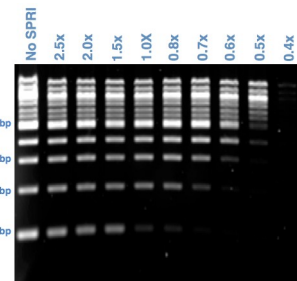


Bad Sample

B1: RNL_165_1



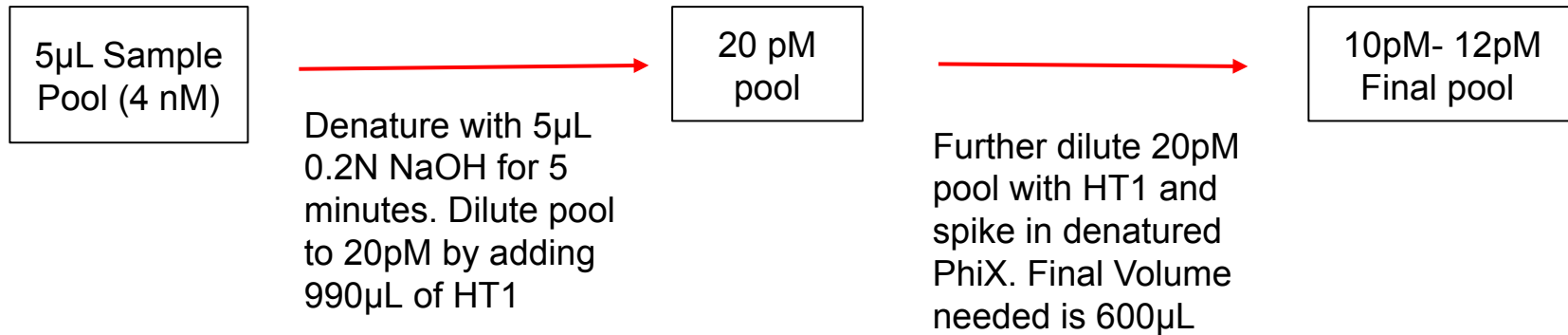
- Average fragment size should be 300-600 bp
- Fragment that is 300 base pairs long has about 150 base pairs of adapter sequences
 - Long enough to run up to PE 150
- Adapter Dimers (~140 bp) are bad because they preferentially cluster to the flowcell
 - Results in inaccurate loading
 - A small percent of dimers will take up a significant proportion of reads.



SPRI Select

Denature and dilute library

- Quantify sample pool by bioanalyzer/tapestation and qPCR (if possible)



- Loading concentration can be variable between MiSeqs (keep a log)
- 1-5% PhiX recommended (15-20% for amplicons)



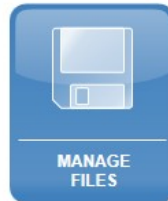
Welcome to Illumina MiSeq

AAAGAAATGATAACAGTAACACACTTCTGTAAACCTTAAGATTACTTG
AATCAACGTAACGTAACGAAAGTATCATTAAGATTACTTGATOCACI
AACGACGAAAAGAAATGATAACAGTAACACACTTCTGTAAACCTTAA
TTAACGTACCATTAAGAGCTACCGTGCAACAGTAACACACTTCTGT
AAAGAAATGATAACAGTAACACACTTCTGTAAACCTTAAGATTACTTG
AAGATTACTTGATOCACGATGAAACGTTAAGATTACTTGATOCACI
AAAGTATCAATTACACTAAATATTAAAGTACCATTAAGAGTCTGTTA

SEQUENCE



PERFORM
WASH



MANAGE
FILES



RUN
OPTIONS



MANAGE
INSTRUMENT



7.49 °C



23.15 °C



Illumina MiSeq

BaseSpace
OptionsLoad Flow
CellLoad
Reagents

Review

Pre-Run
Check

Sequence

Post-Run
Wash

Please select BaseSpace option.



Use BaseSpace for storage and analysis

(all raw data from this run will also be stored on this instrument)

Enter your MyIllumina account

Email: angela.detweiler@czbiohub.org



Password:



If you do not have an account, please go to <https://my.illumina.com> to sign up.

Back

Exit

Next

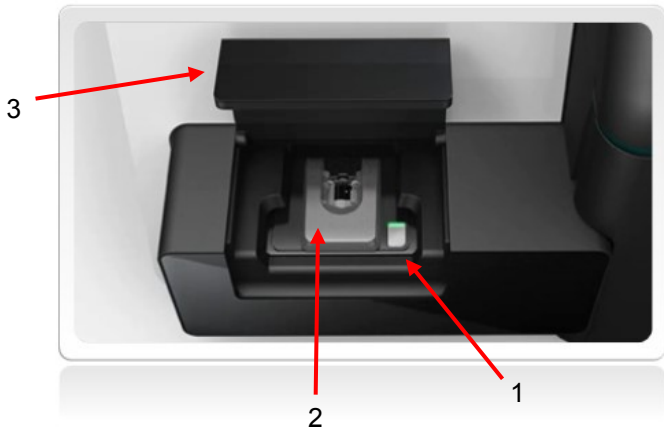
Log into
BaseSpace, then
click Next.



Illumina MiSeq

- BaseSpace Options
- Load Flow Cell
- Load Reagents
- Review
- Pre-Run Check
- Sequence
- Post-Run Wash

Please load the flow cell and wait for the RFID to be read.



Flow Cell Barcode:

- ✓ Flow cell door
- ✓ Clamp engaged
- ✗ Flow cell RFID read



◀ Back

✕ Exit



Press the white release button to open the flow cell clamp (1). Seat the flow cell and close the clamp (2). Close the flow cell compartment door (3). Once all items , click "Next".

Illumina MiSeq

BaseSpace
OptionsLoad Flow
CellLoad
Reagents

Review

Pre-Run
Check

Sequence

Post-Run
Wash

Please load the PR2 and waste bottle.



PR2 Bottle ID: MS9958013-00PR2
Flow Cell Barcode: 000000000-J748W

- ✓ PR2 bottle present
- ✓ PR2 RFID read
- ✓ Waste Bottle Present
- ✓ Waste Bottle Empty



Back



Exit



Next

Remove/dispose of spent reagents and return the empty waste container. Remove the wash PR2 bottle and load the new buffer. Once all items , click "Next".

*Make sure that the sipper handle is down after inserting the PR2 and waste bottle. This will also come handy when troubleshooting.

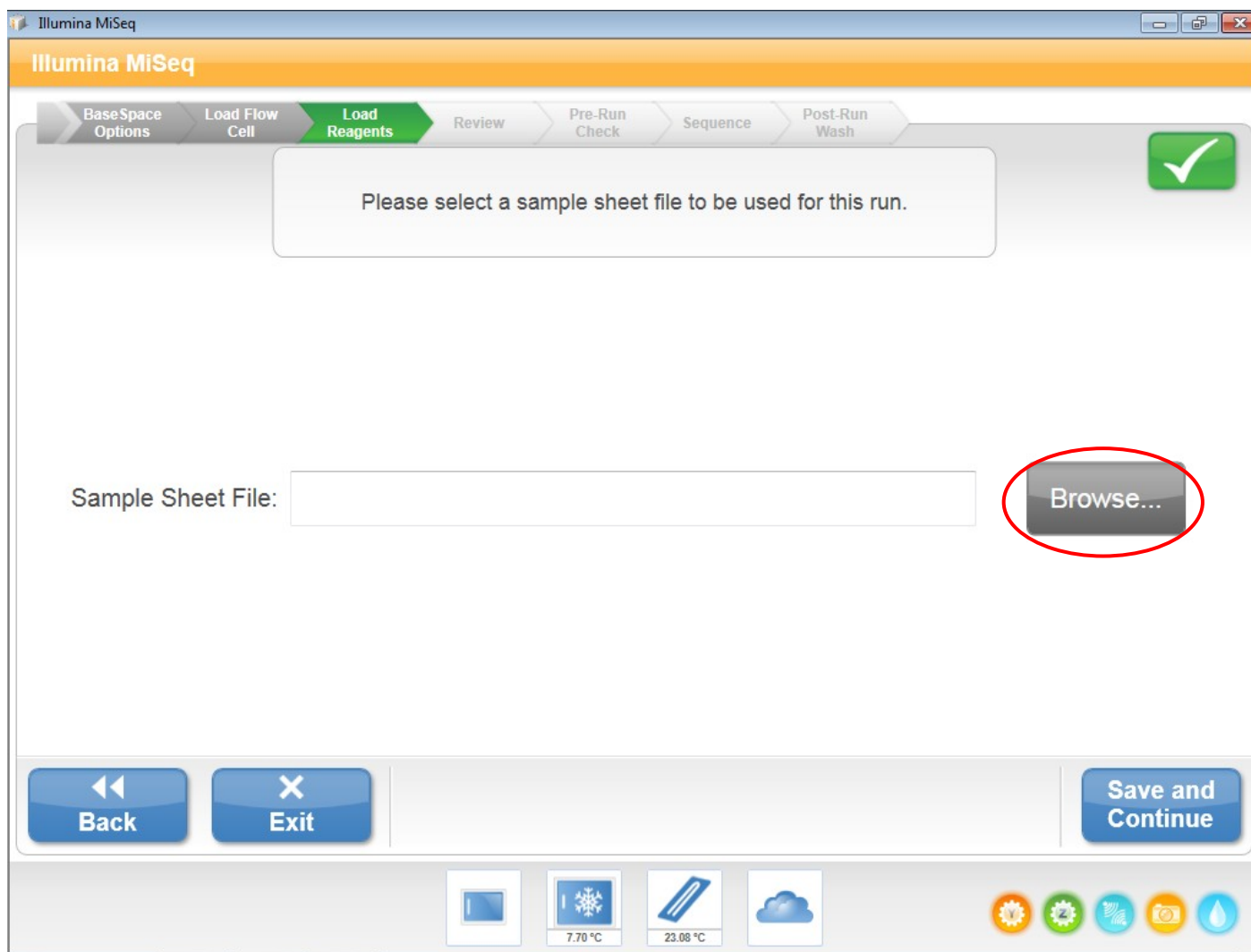


7.51 °C



23.04 °C





The MiSeq requires that the user upload a sample sheet. At CZ Biohub, we download the sample sheet from gDrive to the Sequencer Desktop. The sample sheet name needs to be changed to the reagent cartridge ID (e.g. "MS3046361-600V3").

Browse for the location of the file.

Samplesheet

20200723_Veronica_Escalante_CZBSeqSub03383 - Default Indices.csv — Edited

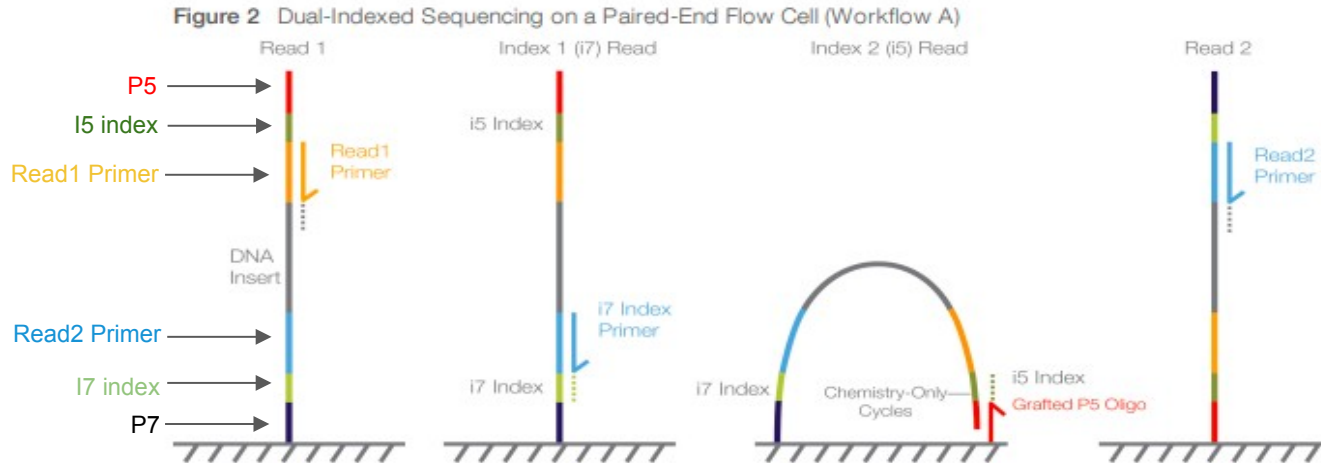
```
[Header],,,,,,,,,,
Date,2020-07-23,,,,,,,,,
Experiment Name, 20200723_Veronica_E,,,,,,,,,
Workflow,GenerateFASTQ,,,,,,,,,
Application,FASTQ Only,,,,,,,,,
Chemistry,Amplicon,,,,,,,,,
IEMFileVersion,4,,,,,,,,,
,,,,,,,,,
[Reads],,,,,,,,,,
250,,,,,,,,,
250,,,,,,,,,
,,,,,,,,,
[CZBiohubSequencing],,,,,,,,,,
Submitter Name,Veronica Escalante,,,,,,,,,
PI Name,Peter Turnbaugh,,,,,,,,,
Task Code,4159,,,,,,,,,
Submission ID,CZBSeqSub03383,,,,,,,,,
Library Pool Name,SPF_SIM_RSV_VE_1,,,,,,,,,
Platform,MiSeq V3,,,,,,,,,
Index 1 Length,8,,,,,,,,,
Index 2 Length,8,,,,,,,,,
Library Number,192,,,,,,,,,
Sample Sheet Status,Form Completed ,,,,,,,,,,
,,,,,,,,,
[Data],,,,,,,,,,
Study_ID,Study_Description,BioSample_ID,BioSample_Description,Sample_ID,Sample_Name
,Sample_Owner,Index_ID,Index,Index2_ID,Index2,Organism,Host,Sex,Tissue_Source
,Understanding how statins shape the microbiome ,,Understanding how statins shape
```

Make sure to add “Experiment Name, XXXX,,,,,,,,” in line 3 as highlighted in the sample sheet example here. This will name the run on BaseSpace.

[Sample sheet example](#)

Index Reads

- When preparing a sample sheet for a MiSeq run-
 - I7 index (Index 1) sequence is the reverse complement of the ordered sequence
 - I5 index sequence (Index 2) is the sequence that you ordered



Illumina MiSeq

Illumina MiSeq

BaseSpace Options Load Flow Cell **Load Reagents** Review Pre-Run Check Sequence Post-Run Wash

Please select a sample sheet file to be used for this run.

Sample Sheet File: MS3046361-600V3

Browse...

Back Exit Save and Continue

Once the sample sheet is selected, click "Save and Continue"

Illumina MiSeq

BaseSpace
Options

Load Flow
Cell

Load
Reagents

Review

Pre-Run
Check

Sequence

Post-Run
Wash



Please load the reagent cartridge.



Reagent ID: MS3046361-600V3
PR2 Bottle ID: MS9958013-00PR2
Flow Cell Barcode: 000000000-J748W

- ✓ Reagent cartridge present
- ✓ Reagent cartridge RFID read
- ✓ Chiller Door Closed

Change
Sample Sheet

Enter Reagent
Kit Barcode



Back



Exit



Next

Finally remove the the wash cartridge, and slide in the new reagent cartridge. Close the reagent cartridge doors. Once all items , click "Next".

Illumina MiSeq

BaseSpace Options Load Flow Cell Load Reagents **Review** Pre-Run Check Sequence Post-Run Wash



Please review the run setup.

Experiment Name: 20200731Veronica_E

Sample Sheet Name: MS3046361-600V3.csv

Analysis Workflow: Generate FASTQ

Chemistry: Amplicon

Read Length:

Read 1	Index 1	Index 2	Read 2
250	8	8	250

BaseSpace User: angela.detweiler@czbiohub.org

Recipe Folder: D:\Illumina\MiSeq Control Software\CustomRecipe
Sample Sheet Folder: C:\Users\misesq01\Desktop
Manifest Folder: D:\Illumina\MiSeq Control Software\Manifests
Output Folder: M:\MiSeq-01

Change Folders

Back

Exit

Next

Review the run setup information pulled from the uploaded sample sheet.

If the Read or Index length needs to be updated, go back to the sample sheet, make changes and re-upload the sample sheet.

Click "Next".

Illumina MiSeq

BaseSpace
OptionsLoad Flow
CellLoad
Reagents

Review

Pre-Run
Check

Sequence

Post-Run
Wash

Checking flow...



Network Connection Active



Flow Cell Cartridge Loaded



Waste Bottle Loaded



Reagent Cartridge Loaded



Chiller Door Closed



Primary Analysis Ready



Sample Sheet Present



Free Disk Space (22%)



PR2 Bottle Loaded



Flow Cell Door Closed

Flow Rate Measured

Reagent ID: MS3046361-600V3
PR2 Bottle ID: MS9958013-00PR2
Flow Cell Barcode: 000000000-J748W



Back



Exit



Start Run

The pre-run check will take a few minutes. Once all items , click "Start Run".

Illumina MiSeq

BaseSpace
OptionsLoad Flow
CellLoad
Reagents

Review

Pre-Run
Check

Sequence

Post-Run
Wash

#MS3046361-600V3 "20200731_Veronica_E"

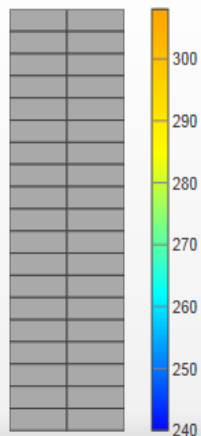
Workflow: Generate FASTQ

250 | 8 | 8 | 250

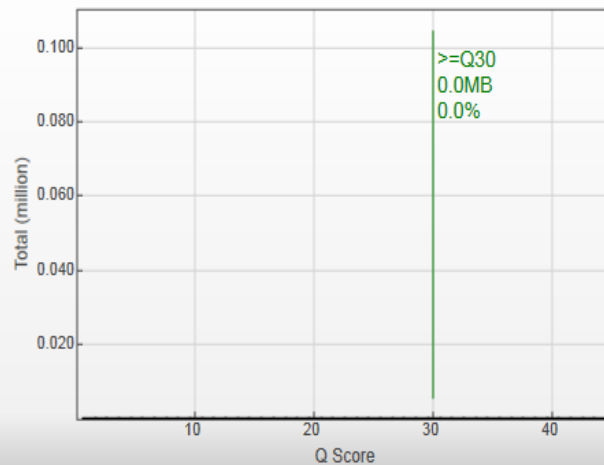
BaseSpace User: angela.detweiler@czbiohub.org

Step 0 of 23: - Cycle 1 of 516

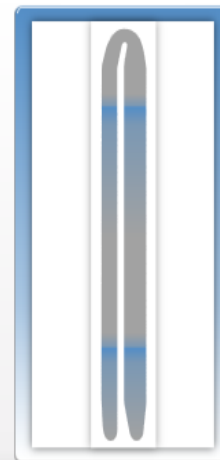
Intensity



Q-Score All Cycles



Flow Cell



Cluster density and PF% metrics will become available after 20 and 25 cycles, respectively.

MiSeq runs can last from 4-56 hrs depending on the kit.

Stop

Pause

Values updated as sequencing progresses

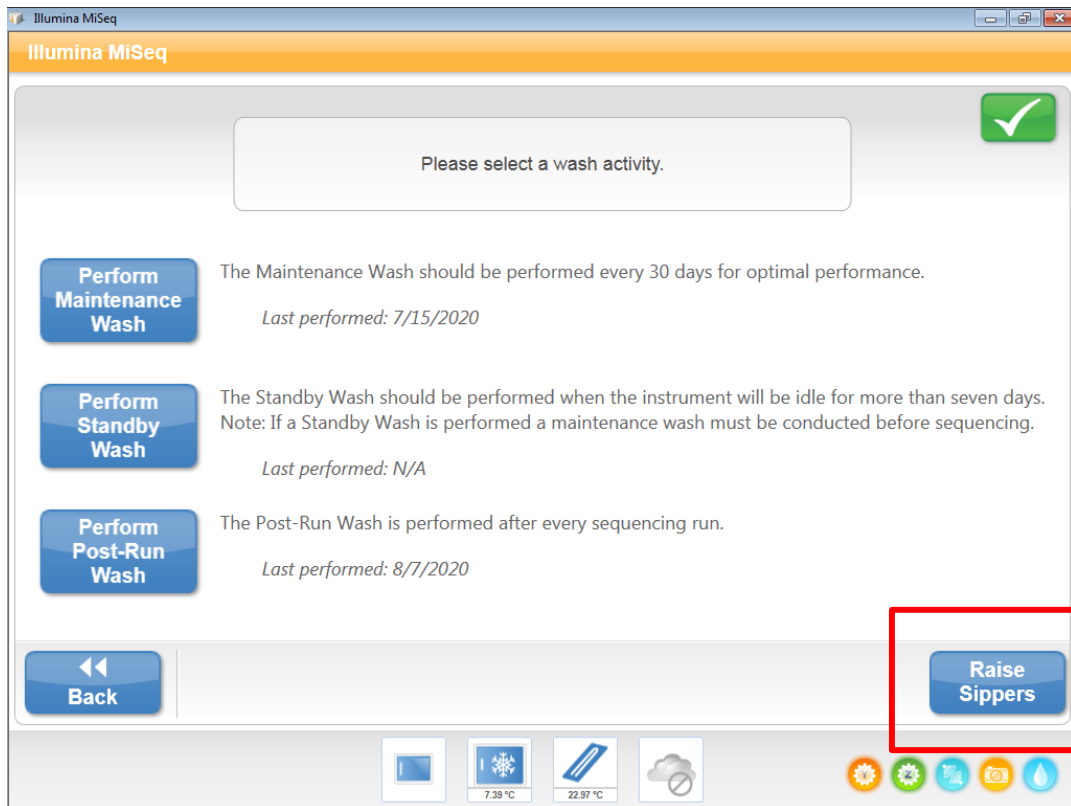
Cluster Density -- K/mm2

Clusters Passing Filter -- %

Estimated Yield -- MB

Perform Wash

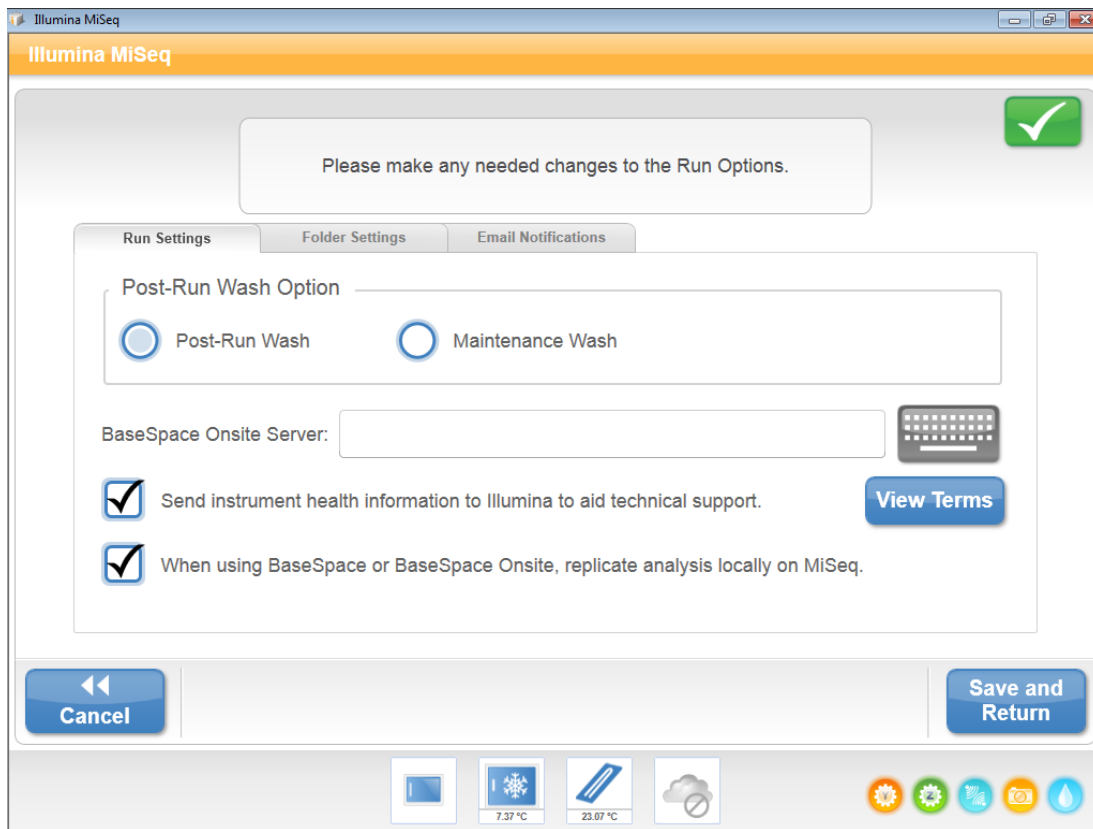
Wash Types



- **Maintenance Wash:** Once every month (~90mins)
 - A series of three wash steps with 0.5% Tween20
- **Stand by Wash:** Only if machine will be idle for for more than a week.
 - A series for two washes with 0.5% Tween20
- **Post-Run Wash:** Mandatory after every run with 0.5% Tween20

When troubleshooting, you may need to raise the sippers manually

Run Options



Illumina MiSeq

Please make any needed changes to the Run Options.

Run Settings | Folder Settings | Email Notifications

Post-Run Wash Option

☒ Post-Run Wash ☐ Maintenance Wash

BaseSpace Onsite Server:

☒ Send instrument health information to Illumina to aid technical support.

☒ When using BaseSpace or BaseSpace Onsite, replicate analysis locally on MiSeq.

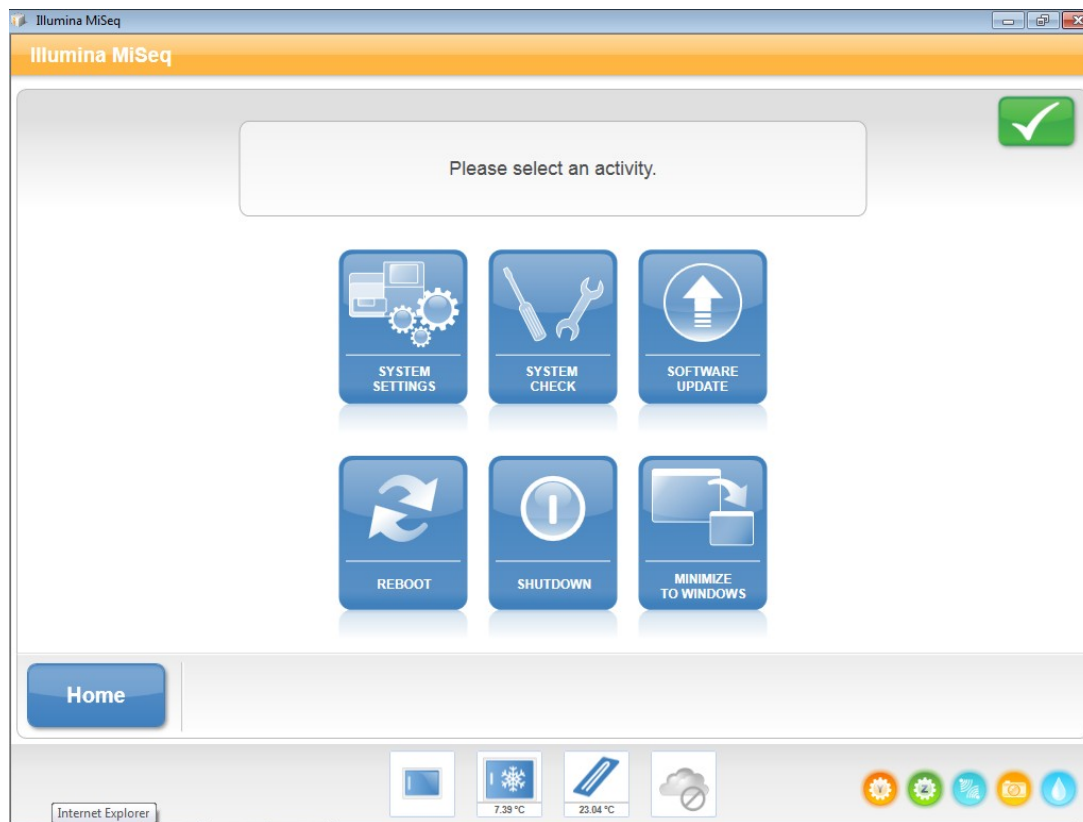
[View Terms](#)

[Cancel](#) [Save and Return](#)

7.37 °C 23.07 °C

The image shows a screenshot of the Illumina MiSeq Run Options window. The window has a title bar with the text 'Illumina MiSeq'. Below the title bar is a header bar with the text 'Illumina MiSeq'. The main content area contains a message box that says 'Please make any needed changes to the Run Options.' with a green checkmark icon. Below this is a tabbed interface with three tabs: 'Run Settings', 'Folder Settings', and 'Email Notifications'. The 'Run Settings' tab is selected. Under this tab, there is a section titled 'Post-Run Wash Option' with two radio buttons: 'Post-Run Wash' (which is selected) and 'Maintenance Wash'. Below this is a text input field labeled 'BaseSpace Onsite Server:' followed by a keyboard icon. There are two checkboxes: 'Send instrument health information to Illumina to aid technical support.' (which is checked) and 'When using BaseSpace or BaseSpace Onsite, replicate analysis locally on MiSeq.' (which is checked). A 'View Terms' button is located to the right of the second checkbox. At the bottom of the window, there are two buttons: 'Cancel' and 'Save and Return'. The bottom of the window also features a status bar with icons for a folder, a snowflake (7.37 °C), a pencil (23.07 °C), a cloud, and several colored circular icons.

Manage Instrument



MiSeq Installation

- For MiSeq installation instructions contact your Illumina Representative.
- For instrument calibration you will need an optical flow cell to run the “Optics Test” in System Check.
- MySeq System Prep Guide - https://support.illumina.com/content/dam/illumina-support/documents/documentation/system_documentation/miseq/miseq-site-prep-guide-15027615-f.pdf
- MiSeq System Guide - https://support.illumina.com/content/dam/illumina-support/documents/documentation/system_documentation/miseq/miseq-system-guide-for-miseq-reporter-1000000061014-00.pdf
- MiSeq Denaturation Guide - https://support.illumina.com/content/dam/illumina-support/documents/documentation/system_documentation/miseq/miseq-denature-dilute-libraries-guide-15039740-10.pdf